INCH-POUND

MIL-H-51290B(EA)
17 May 1991
SUPERSEDING
MIL-H-51290A(EA)
18 December 1989

MILITARY SPECIFICATION

HOOD, AIRCRAFT CHEMICAL-BIOLOGICAL MASK: M7

This specification is approved for use by the U.S. Army *Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.*

1. SCOPE

1.1 Scope. This specification covers one type and one size of protective hood.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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SPECIFICATIONS

FEDERAL

U-T-30 - Talcum Powder
MMM-A-1617 – Adhesive, Rubber Base, General Purpose

MILITARY

MIL-C-51251 - Cloth, Coated; CBR Protective

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 101 - Test Procedures for Packaging Materials
FED. TEST METHOD STD. NO. 191 - Textile Test Methods
FED-STD-595 - Colors

(’Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 191 11-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-35-73 - Hood, Aircraft Chemical-Biological Mask: M7

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-35-73 - Hood, Aircraft Chemical-Biological Mask, M7

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CT, Rock Island, IL 61229-7210.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation.

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Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

ASTM STANDARDS

D 1579 - Filler Content of Phenol, Resorcinol, and Melamine Adhesives

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-35-73 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.1.1 Elastic cord. The elastic cord for the lens opening shall be a braided cotton elastic .-cord, 3/16 $\pm$ 1/32 inch in diameter, OG 107, weighing no less than 0.18 ounce per linear yard when tested as specified in 4.3.4.1.

3.1.1.1.1 Core. The core shall be composed of 7 ends of 36-gage, natural or synthetic rubber. wrapped first with four ends of 20/1 and on top with one end of 20/1 cotton yam when tested as specified in 4.3.4.1.

3.1.1.1.2 Covering. The covering shall be made with 16 carriers, 1 end per earner, with 2 ends braiding as one; or 8 carriers, 2 ends per carrier. The cover shall be braided with a basket weave, with no less than 26 picks per inch, using 20/2 cotton yam when tested as specified in 4.3.4.1.

3.1.1.1.3 Initial elongation and permanent set. The cord shall have an initial elongation, with a 32-ounce weight, of 130 $\pm$ 10 percent and an initial permanent set of no more than 15 percent when tested as specified in 4.3.4.1.1 and 4.3.4.1.2.

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3.1.1.1.4 Accelerated aging. After aging as specified in 4.3.4.1.4, the change in elongation shall be not greater than 25 percent and permanent set shall be not greater than 25 percent when tested as specified in 4.3.4.1.1 and 4.3.4.1.2. The cord shall show no rupture of elastic strands after aging.

3.1.1.1.5 Low temperature elongation. The cord shall retain no less than 20 percent of its initial elongation when tested as specified in 4.3.4.1.3.

3.1.2 Components. All components of the hood shall conform to the specifications listed on Drawing 5-35-73 and on subsidiary drawings.

3.1.3 Pattern configuration. The configuration of the cut materials shall meet the applicable pattern drawing when examined thereto.

3.2 Assembly. The hood shall be assembled in accordance with Drawing 5-35-73.

3.2.1 Stitching. All rows of stitching shall be regular lines free from loose stitching or thread loops. All stitching shall be of continuous thread. All seams shall be backstitched at the ends with a minimum of three stitches. All loose thread ends shall be trimmed to 1/4 inch or less.

3.2.2 Marking. The finished hood assembly shall be legibly and indelibly marked as specified on Drawing 5–35-73.

3.3 Fastening devices. The pile and hook fastener, drawcord, drawcord clip, lens opening and buckles shall function smoothly and efficiently.

3.4 Light penetration. When the hood is examined over a 100-watt light as specified in -4.4.6.1, there shall be no light penetration through any hole in the body or through six or more holes in the seams of the hood. However, the seams around the lens opening in the front and the hem along the bottom of the hood shall be exempt.

3.4.1 Windows (film transparencies). Windows are permissible provided that no leakage is indicated when tested as specified in 4.4.6.1. Areas that show leakage shall be considered as holes.

3.4.2 Luting compound. A luting compound shall be used on the seams of the hood. The luting compound shall be a self-curing solvent neoprene rubber diisocyanate adhesive (see 6.3) and shall be applied on the inside of the hood. When used in combination with the coated fabric, the luting compound shall meet the following requirements when tested as specified in 4.3.4.2:

Blocking of seams	Scale rating No. 3
Resistance to low temperature	Pass test with no cracking, flaking, or

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	separation from the fabric
Stiffness (centimeters)	
Initial	10 maximum
*After heat aging	12 maximum
Solids content	20 percent minimum

* The specimen shall show no cracking, flaking, or separation from the fabric.

3.4.3 Color. The color of the finished hood shall be olive drab and be within the range of FED-STD-595 color chip numbers 34086 through 34102.

3.4.4 Dusting. After the luted seams have thoroughly dried, they shall be dusted with talcum powder conforming to U-T-30.

3.5 Seam configuration. The configuration of each seam shall comply with the applicable section views of Drawing 5-35-73 when inspected as specified in 4.4.6.2.

3.6 Repairs.

3.6.1 General repairs. Repairs are permitted on the hood for cuts, nicks (1/4-inch maximum length), or pinhole punctures greater than l/4-inch from seam stitching. Repairs shall be accomplished by securely cementing an approximate 1-l/2-inch diameter patch of material conforming to MIL-C-51251 to the inside of the hood so as to completely cover the defective area. The cement shall conform to type 1 or type II of MMM-A-1617. No more than one such repair shall be permitted on any single hood panel.

3.6.2 Pinhole repairs. Pinholes within l/4-inch of a seam stitching shall be repaired by . applying luting compound and repaired area must meet light penetration requirement (see 3.4).

3.7 First article. Unless otherwise specified in the contract, a first article sample shall be produced in accordance with this specification for examination and tests (see 4.3 and 6.2), prior to the start of regular production.

3.8 Workmanship. The hood shall be free from damage such as tears, rips, cuts, nicks, abrasions or punctures and contamination such as dirt, stains, grease or oil. All seams shall be free from pleating and distortion.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet-all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to Applying such alternative procedures, methods, or- equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.4.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. AH approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1. Inspections to be performed. As determined by the Government, the sample first article items shall be examined and tested in accordance with this specification using either the first article procedures in 4.3.3 or other first article procedures which will be provided by

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the Government. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings.

4.3.2 Sample. The first article sample, manufactured and packaged (when military packaging is required) using the same methods, materials, equipment, and -processes as will be used during regular production, shall consist of the following: 25 hoods, 3 sets of pattern blanks, 1 sample seam (4.3.4.2.1), 1 pint of luting compound, a 10 foot long sample of elastic cord and 3 empty unit pack containers. Empty unit pack containers shall not be part of the first article sample when packaging is for interplant shipment. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.3 Inspection procedures. Each hood in the first article sample shall be examined and tested for compliance with all the requirements of the applicable drawings, special packaging instructions and this specification (unless otherwise specified in table I). Hoods selected at random, pattern blanks, the sample seam, luting compound and the empty unit pack containers from the first article sample shall be subjected to the tests specified in table 1.

TABLE I. First article inspection procedures

Inspection procedure	Number of first article samples
Pattern configuration (4-3.3. 1)	3 sets of pattern blanks
Elastic cord (4.3.4.1)	Sample of elastic cord
Luting compound (4.3.4.2)	1 sample seam
Seam configuration (4.4.6.2)	5 hoods
Unit pack container leakage (4.4.7.1)	5 unit packed hoods
Unit pack container seam (4.4.7.2)	3 empty unit pack containers

4.3.3.1 Pattern configuration. The first article sample of pattern blanks shall be examined for compliance with the pattern drawings cited on Drawing 5-35-73.

4.3.4 Tests.

4.3.4.1 Elastic cord, core, and covering. The methods of testing specified in FED. TEST METHOD STD. NO. 191, wherever applicable and as listed in table II, shall be followed. All requirements are applicable to the sample unit.

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TABLE II. Test methods for elastic cord. core and covering

Component	Characteristic	Test method	Specimens	Reported results
Elastic cord (3.1.1.1)	Weight	Standard	3	To nearest 0.01 ounce
	Diameter	Standard	1	To nearest 1/32 inch
	Elongation	Standard	1	To nearest 1.0 percent
Core (3.1.1.1.1)	Gage of rubber	Standard	1	To nearest gage
	Yarn for wrapping (ply, size and number of yarns)	Count	1	Yarn size
	Number of ends	Count	1	To nearest whole number
Covering (3.1.1.1.2)	Number of carriers	Count	1	To nearest whole number
	Ends per carrier	Count	1	To nearest whole number
	Picks per inch	Count	1	To nearest whole number
	Yarn (size and ply)	Count	1	Yarn size

4.3.4.1.1 Elongation. Elongation shall be determined under laboratory conditions of 65 ± 2 percent relative humidity, and a temperature of 70° to 80°F. A 14-inch specimen shall be cut from the sample of cord, and two marks shall be made on the specimen so that a distance of 10 inches shall be between the gage marks. The specimen shall be suspended from a clamp in such a manner as to allow a weight to be hung on the lower end. The weight shall be gradually lowered until the entire load is carried by the specimen. After 2 minutes, a measurement shall be taken between the two marks and the increase in length calculated as percent elongation.

4.3.4.1.2 Permanent set. The 14-inch test specimen shall be marked so that a distance of 10 inches is between the gage marks and then shall be stretched 50 percent and held there for 24 hours under laboratory conditions of 65 ± 2 percent relative humidity and a temperature of 70° to 80°F; the specimen shall then be released and allowed to rest for 10 minutes. The extension remaining shall be measured and expressed as a percentage of the original distance between the gage marks.

4.3.4.1.3 Low temperature elongation. A 14-inch test specimen of cord shall be marked with two marks so that a distance of 10 inches is between gage marks. The specimen shall

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then be exposed to a temperature of -40° ± 5°F for two hours after which, at that temperature, it shall be tested for elongation in accordance with 4.3.4.1.1 under an elongation load.

4.3.4.1.4 Accelerated aging (oxygen bomb). The test specimens selected for accelerated aging in the presence of oxygen shall be suspended for a continuous period of 96 hours in a closed chamber and subjected to oxygen having a pressure of 275 to 300 pounds per square inch and maintained at a temperature of 1580 ± 2°F. At the end of the 96 hour period, the test specimens shall be removed from the chamber and allowed to stand 24 hours and then tested as specified in 4.3.4.1.1 and 4.3.4.1.2.

4.3.4.2 Luting compound.

4.3.4.2.1 Sample seams. Approximately 2-1/8 yards of sample seam, 7 inches wide, shall be prepared by joining two pieces of the coated cloth with the type of stitch and seam defined in Section C-C of Drawing 5-35-73. The seam shall be coated with Ming compound (see 3.4.2).

4.3.4.2.2 Blocking resistance. Resistance to blocking of the luting compound shall be tested on a sample prepared as specified in 4.3.4.2.1. Test method 5872 of FED. TEST METHOD STD. NO. 191 shall be used with the following exceptions:

(a) Test specimens shall measure approximately 6 by 8 inches with the seam in the center running the full 8-inch length.

(b) The seam shall be folded over on itself once to a size approximately 6 by 4 inches so that the luting compound is on the inside of the folded specimen.

(c) The test temperature shall be 71.1° ± 1.0°C.

(d) A 1 ± 1/16 pound weight shall be distributed evenly along the seam.

4.3.4.2.3 Low temperature resistance. The resistance of the luting compound to low temperature shall be tested on a sample prepared as specified in 4.3.4.2.1. Test method 5874 of FED. TEST METHOD STD. NO. 191 shall be used with the following exceptions:

(a) Test specimens shall measure approximately 6 by 8 inches with the seam in the center running the full 8-inch length.

(b) The seam shall be folded over on itself to a size of approximately 6 by 4 inches so that the luting compound is on the inside of the folded specimen.

(c) The roller weight shall be run along the fold line.

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4.3.4.2.4 Stiffness.

(a) Initial. A two-mil film (dry thickness) of the luting compound, approximately 7 by 13 inches, shall be cast on the coated cloth. The film shall be allowed to air dry for a minimum of 24 hours. Test specimens shall be cut from the above prepared cloth and the stiffness shall be determined as specified in method 5204 of FED. TEST METHOD STD. NO. 191.

(b) After heat aging. Test specimens shall be prepared as specified in 4.3.4.2.1 and then placed in a circulating air oven at 82.2° ± 1.0°C for 24 hours. After the specimens are removed from the oven, the y shall be allowed to cool for a minimum of 1 hour and then the stiffness determined as specified in method 5204 of FED. TEST METHOD STD. NO. 191.

4.3.4.2.5 Solids content. The solids content of the luting compound shall be determined as specified in ASTM Standard D 1579.

4.3.5 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the hoods produced by one manufacturer, at one plant, from the same materials using one color of coated-cloth, under essentially the same manufacturing conditions, and at essentially the same time. However, no more than five consecutive lots of butyl coated cloth from one manufacturer shall be represented in any one lot of finished hoods; the strap, binder strip and reinforcing patch components need not be considered in the application of this lotting criteria.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.5 and, when specified, table III. Samples shall be selected at random.

4.4.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

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4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

TABLE III. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	l	l	l	5	3	2
9 to 15	l	*	l	*	*	*	13	8	5	3	2
16 to 25	*	*	l	*	l	20	13	8	5	3	3
26 to 50	*	l	*	*	32	20	13	8	5	5	5
51 to 90	l	*	*	50	32	20	13	8	7	6	5
91 to 150	*	l	125	50	32	20	13	12	11	7	6
151 to 280	l	l	125	50	32	20	20	19	13	10	7
281 to 500	l	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
.3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

* Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.4.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table III for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER
4.4.5(a)	Hood, aircraft chemical-biological mask, M7 SHEET 1 OF 3			5-35-73
				NEXT HIGHER ASSY
CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
critical	None defined			
Major				
101	Light penetration through hood	Table III, level VI	3.4	4.4.6.1
102	Light penetration through needle holes in scams	Table III, level VI	3.4	4.4.6.1
103	Heavy coated side of cloth on exterior of hood	Table III, level VII	3.2	VI
104	Component match marks misaligned no more than l/8-inch	Table III, level VII	3.2	CE
105	Seam configuration	5; Acc 0, Rej 1	3.5	4.4.6.2*
106	Specified components correctly assembled	Table III, level VII	3.1.2	VI
107	Fastening devices (hook and pile fastener, drawcord, buckle, lens opening and drawcord clip) correctly function	Table III, level VII	3.3	VI
108	Stitching not run off	Table III, level VII	3.2.1	VI and CE
1.09	Number of stitches per inch correct	Table III, level VII	3.2	CE
110	Dusting of luted seams with talcum powder sufficient to prevent tackiness	Table III, level VII	3.4.4	VI
111	Marking correct	Table III, level VII	3.2.2	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection
* Destructive test

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER
4.4.5(a)	Hood, aircraft chemical-biological mask, M7	SHEET 2 OF 3		5-35-73 NEXT HIGHER ASSY
CATEGORY	CHARACTERISTIC	SAMPLING AND Acceptance CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
112	Workmanship other than 201 and 208	Table XII, level VII	3.8	VI
113	Hood material double coated	Table III, level VII	3.1.1	VI
114	No uncoated areas	Table III, level VII	3.1.1	VI
115	No separation of butyl rubber from base cloth	Table III, level VII	3.1.1	VI
116	No brittle areas that crack when bent	Table III, level VII	3.1.1	VI
117	No embedded foreign material	Table III, level VII	3.1.1	VI
118	Pattern configuration	Note 1	3.1.3	CE
Minor				
201	No distortion or pleating	Table III, level XI	3.8	VI
202	Stitch line broken less than l/4-inch	Table III, level XI	3.2.1	VI and CE
203	Stitching tension correct (not loose, causing exposed loops, or tight, causing pleating)	Table III, level XI	3.2	VI
204	Stitching margins as specified	Table III, level XI	3.2	CE
205	Thread ends trimmed to l/4-inch or less	Table III, level Xl	3.2.1	CE
206	Seam backstitched at least three stitches (when not held by another row of stitches)	Table III, level XI	3.2.1	VI
207	Stitch type and pattern as specified	Table III, level XI	3.2	VI and CE

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection
Note 1. A sample of 5 die-cut blank shall be taken from each lot. When blanks arc knife-cut, two samples shall be taken from each blank cutting, one from the top and one from the bottom. Rejection criteria shall be accept on 0 and reject on 1.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.5(a)	Hood, aircraft chemical-biological mask, M7	SHEET 3 OF 3	5-35-73
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
208	No minor foreign matter (minor foreign matter is small areas of stain, dirt, or machine lubricant (other than oil or grease) not exceeding 1/4 square inch area and a total area of foreign matter not exceeding 1 square inch)	Table III, level XI	3.8	VI and CE
209	Marking legible and properly located	Table III, level XI	3.2.2	VI
210	Repairs as specified	Table III, level XI	3.6	VI and CE
211	No area of excessive coating (solid runs, ridges or pimples)	Table HI, level XI	3.2	VI
212	No tacky coating	Table HI, level XI	3.2	VI
213	Colors and shades as specified	Table Ill, level XI	3.4.3	VI*

NOTES:

CE - Commercial inspection equipment
VI - Visual inspection
* Before making color matchings, the dusting powder shall be removed from the hood by wiping with a damp cloth.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.5(b)	Packaging	SHEET 1 OF 1	SPI P5-35-73 NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Unit pack container leakage	Table III, level X	5.2	4.4.7.1
102	Unit pack container seam	5; Acc 0, Rej 1 *	5.2	4.4.7.2**
103	Hood completely clean and dry prior to unit packing	Table III, level x	5.2	VI
104	Talcum evident and correct	Table III, level x	5.2	VI
105	Unit pack container evident and correct	Table III, level x	5.2	VI and CE
106	Unit pack container markings evident, correct, and legible	Table III, level x	5.2	VI and CE

NOTES:
This paragraph is not applicable to packaging for interplant shipment.
* Samples shall consist of closed empty unit pack containers.
**Destructive test
CE - Commercial inspection equipment
VI - Visual inspection

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4.4.6 Tests.

4.4.6.1 Light penetration. The hood shall be examined over the specified light at rated voltage for light penetration. The light shall be no farther than 12 inches from the area examined and the observer no farther than 24 inches. Each seam shall be examined from at least two angles. The light intensity of the area around the hood being examined shall be no more than 32 footcandles. Windows or light areas which are suspected of containing holes shall be subjected to a hydrostatic pressure of 2 pounds per square inch minimum for 60 $\pm$ 5 seconds and observed for evidence of water leakage.

4.4.6.2 Seam configuration. The hood seams shall be sectioned as shown on Drawing 5-35-73 and inspected for compliance thereto.

4.4.7 Packaging tests.

4.4.7.1 Unit pack container leakage. The unit pack container and its contents, unit packed for shipment, shall be tested for leakage in accordance with method 5009.3 of FED. TEST METHOD STD. NO. 101 using the hot water technique.

4.4.7.2 Unit pack container seam. The closure seam of the empty unit pack container shall be prepared under the same conditions used for unit pack assemblies. The closure seam of the empty unit pack container shall be tested in accordance with method 2024 of FED. TEST METHOD STD. NO. 101, except that the static load shall be 38 (+2 or -0) ounces and a 5 percent reduction in the static load shall be permitted when the room temperature exceeds 90°F in the test area.

5. PACKAGING

5.1 Interplant shipment (see 6.5). Military packaging as specified in 5.2 is not applicable for interplant shipments. The hoods shall be packaged to provide adequate protection from physical damage and marked to identify the hoods and provide any other required data during shipment from the source to the first receiving activity.

5.2 Repair or replacement parts. For all shipments, other than interplant shipments (see 6.5), packaging (preservation, unit packing, packing and marking) shall be as specified in SPI P5-35-73.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M7 hood is intended for use with the M24 chemical-biological aircraft mask.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)
(c) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(d) Whether military packaging or interplant shipment is required (see Section 5)
(e) Level of unit packing required (see 5.2)
(f) Level of packing required (see 5.2)
(g) Shelf life code marking

6.3 Luting compound. The following compound has been used satisfactorily and may be used provided it meets the requirements of 3.4.2.

PART I	*PART II
100.0 parts by weight of Neoprene (type WRT)	10 parts by weight of Hylene M-50
4.0 parts by weight of calcined MgO (extra light)	
2.0 parts by weight of Neozone D	
5.0 parts by weight of zinc oxide	
145.0 parts by weight of methylethylketone	*Shall be added to PART I just prior to to use
75.0 parts by weight of Toluene	
2.5 parts by weight of Thiuram E plus dye	

6.4 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.5 Interplant shipment (see 5.1). Packaging and marking for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or between contractors and subcontractors, or from a vendor or contractor to a military arsenal,

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plant or other activity for evaluation, immediate use, or further processing as specified in the applicable contract.

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

butyl-coated cloth
chemical warfare agents
protective clothing
protective equipment
protective masks

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A018

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